The complementary-distribution challenge for phonological learning: The case of [h] and [ŋ] in English

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Abstract

The distribution of [h] and [ŋ] in English poses a well-known challenge for the structuralist assumption that segments in complementary distribution are allophones of the same phoneme. In this paper, we show that this distribution also poses a challenge within generative linguistics, a challenge of explanatory adequacy. The challenge is to explain why the English-learning child does not acquire a grammar in which [h] and [ŋ] are derived from the same underlying representation. The generative literature has not identified this as a problem, largely because of the assumption that [h] and [ŋ] are two phonetically dissimilar to participate in an alternation. However, we argue that phonetic accounts do not eliminate the challenge. We discuss simplicity-based theories of learning and show that under naive representational assumptions, simplicity-based learners prefer to derive [ŋ] from an underlying /h/, rather than a longer sequence of /ng/ (or an underlying /ŋ/), and therefore make wrong predictions about English speakers' knowledge. The same learners, however, make correct predictions when combined with a representational framework that assumes underspecification and lexical redundancy rules. We therefore conclude that the acquisition problem can be addressed using simplicity-based learning, and that the complementary distribution of [h] and [ŋ] provides an argument for underspecification and lexical redundancy rules.

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1 Introduction

The consonants [h] and [ŋ] are in complementary distribution in English, a fact that was identified as a puzzle for structuralist accounts as early as Twaddell (1935, p. 29). He writes:

> A perfectly consistent and unambiguous transcription of English is possible with the use of a single symbol in the place of [h] and [ŋ]; but no phonetician or phonologist could consent to regarding these two 'sounds' as variants of the same phoneme.

This observation poses a well-known challenge for the structuralist phonemic method, according to which segments in complementary distribution belong to the same phoneme. This challenge has long been set aside under the assumption that alternating segments must be phonetically similar, unlike [h] and [ŋ]; yet what would count as a sufficiently concrete, phonetic account that rules out their alternation is not straightforward (Hyman, 1975, p. 64). As we will show, the distribution of [h] and [ŋ] poses a further, underappreciated challenge – this time for generative phonology, and specifically for phonological learning. Moreover, and contrary to the prevailing assumption, this challenge is unlikely to be resolved by appeals to phonetic similarity.

We will spell out this challenge in sections 2 and 3 but for now, simplifying, there are at least three observationally adequate grammars of English that are consistent with the distribution of [h] and [ŋ]: the descriptively correct grammar of English, which derives [ŋ] from underlying /ng/; a descriptively incorrect grammar that derives [ŋ] from /h/ in coda position; and another incorrect grammar that derives [ŋ] from an underlying /ŋ/. The challenge is to achieve explanatory adequacy with respect to the descriptively correct grammar – that is, to explain why the learner selects the correct grammar over the observationally adequate alternatives.[1]

The challenge is raised by the simpler lexical representations of the incorrect grammars. In those grammars, words like '*song*' can be represented with one fewer segment than their counterparts in the correct grammar (e.g., /sɔ**h**/ or /sɔ**ŋ**/ instead of /sɔ**ng**/). As we will show in section 3, under naive representational assumptions, a family of simplicity-based learners favors the incorrect /h/ → [ŋ] grammar.[2] These are learners that prefer the acquisition of the simplest observationally adequate grammar, where simplicity is assessed over the combined information in the lexicon and the phonological rules or constraints, as in the Sound Pattern of English (SPE; Chomsky and Halle 1968). We will use the term SIMPLICITY, as defined in (1), to refer to all such simplicity-based learning approaches.

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[1] We use Chomsky's (1965) levels of adequacy: '*Observationally adequate*' as a term describing consistency with the data; '*Descriptively adequate*' means that the grammar reflects speakers' knowledge; '*Explanatory adequate*' means that the theory can correctly predict the acquisition of the adult grammar.

[2] As we will discuss later, a solution to the /ŋ/ → [ŋ] challenge exists independently, and we therefore focus mainly on the /h/ → [ŋ] challenge for SIMPLICITY.

(1) SIMPLICITY: A learning theory according to which the learner chooses the simplest observationally adequate grammar, all other things being equal.

The challenge for SIMPLICITY may have gone unnoticed for decades because SIMPLICITY was largely abandoned within phonology. However, SIMPLICITY has recently returned in the form of Minimum Description Length (MDL; Rissanen 1978). MDL is a simplicity-based learner that has been implemented with demonstrated success and, as we will discuss in section 3, has been shown to support the acquisition of phonological grammars and morphological segmentation from distributional evidence (Rasin and Katzir, 2016; Rasin et al., 2021). As will be discussed in section 3, the challenge posed by the /ŋ/ → [ŋ] grammar is already resolved through constraints on underlying representations (CURs). As Rasin and Katzir (2020) show, CURs are essential for phonological learning and, combined with MDL, motivate the reduction of the underlying inventory in a way that rules out underlying /ŋ/. The /h/ → [ŋ] challenge, however, is – to our knowledge – the first empirical challenge for the MDL learner, and we therefore believe that it cannot be ignored. At the same time, the solution we sketch below is the first solution to the explanatory adequacy challenge identified in this paper, and it crucially relies on SIMPLICITY. Whether other learning approaches can account for the same challenge – selecting the correct grammar of English over the alternatives – will remain an open question.

As we will show in section 5, SIMPLICITY's preference of the /h/ → [ŋ] grammar holds only under the assumptions of full specification when combined with CURs. Significantly, however, the simplicity-based learning criterion is not tied to full specification or any other specific representational assumption, and different choices about the architecture of grammar can affect its predictions regarding learning. As we will see, SIMPLICITY makes the correct predictions when combined with Chomsky & Halle's (1968) architecture, which assumes underspecification and lexical redundancy rules – a specific form of CURs in which rules fill in unspecified features in the lexicon. A summary is given in (2).

(2) A summary of our observation from section 5

Representational assumptions	SIMPLICITY's prediction regarding learning
Richness of the Base	English speakers will learn the **/ŋ/ → [ŋ] grammar**
Full specification + CURs	English speakers will learn the **/h/ → [ŋ] grammar**
Underspecification + Lexical redundancy rules (a specific form of CURs)	English speakers will learn the **/ng/ → [ŋ] grammar**

As shown in (2), under Richness of the Base (ROTB; Prince and Smolensky 1993/2004; Smolensky 1996), SIMPLICITY favors the /ŋ/ → [ŋ] grammar. ROTB, however, has been independently rejected on the grounds that, as noted above, CURs are crucial for phonological learning (Rasin and Katzir, 2020). This is why we treat the competition between the /h/ → [ŋ] and the /ng/ → [ŋ] grammars as the primary concern for SIMPLICITY. For other learning approaches, the /ŋ/ → [ŋ] grammar may pose a more significant challenge.

We will elaborate on the observations in (2) in sections 3 and 5, but here is the logic in a nutshell. The incorrect /h/ → [ŋ] grammar includes underlying /h/ in coda position (e.g, /sɔh/ → [sɔŋ] '*song*'), and therefore misses a generalization that otherwise holds for the English lexicon: there are no /h/'s in coda positions whatsoever. Consequently, this grammar must make use of sufficiently many features in the lexicon to distinguish coda /h/ from all other consonants in coda position. As we will show in section 5, this bears on the specification of coda /k/, which is the featurally closest segment to /h/ in English. To remain distinct from /h/, coda /k/ must be specified using an additional feature, such as [±continuant] (distinguishing words like '*song*' and '*sock*' for example), as illustrated in table (3). The correct /ng/ → [ŋ] grammar, by contrast, preserves the generalization that /h/ is excluded from coda position, making the feature [-continuant] redundant for coda /k/, as illustrated in table (4). Under a theory with underspecification and lexical redundancy rules, this feature can be removed from all coda /k/'s across the lexicon.

(3) Underspecified segments in coda position, **incorrect grammar**

	p	b	...	k	h
±coronal	-	-	...	-	-
±anterior	+	+	...	-	-
...			...		
±voice	-	+	...	-	-
±continuant	-	-	...	-	+

(4) Underspecified segments in coda position, **correct grammar**

	p	b	...	k
±coronal	-	-	...	-
±anterior	+	+	...	-
...			...	
±voice	-	+	...	-
±continuant	-	-	...	

As we will see, the elimination of this feature from the lexicon in the correct /ng/ → [ŋ] grammar is more cost-effective than the simplification of lexical representations in the incorrect /h/ → [ŋ] grammar. The reasoning, presented in greater detail in section 5, is as follows. When segments can be underspecified, the representation of coda /k/ requires fewer features in the correct grammar; however, the derivation of [ŋ] is cheaper in the incorrect one, because the representation of /h/, from which [ŋ] is derived, requires fewer features than /ng/. The cost of additional features is aggregated over the number of occurrences across the entire lexicon. Because there are sufficiently more coda [k]'s than [ŋ]'s, SIMPLICITY favors the correct grammar. Seen from this perspective, the complementary-distribution challenge does not truly challenge SIMPLICITY, but rather supports the representational assumptions of the SPE, given in (5). Under these assumptions, segments can be minimally specified in any context across the lexicon, provided that each segment remains uniquely specified. The remaining features, which are redundant for the purpose of distinctness, are filled in by phonological rules.

(5) Representational assumptions that can solve the learning challenge (cf. Chomsky and Halle 1968, pp. 380–389)

1. In every phonological context (e.g., intervocalically, onset, coda, before a strident, etc.), underlying segments can be underspecified for phonological

features as long as they are pairwise distinct.

2. Unspecified features are filled in by lexical redundancy rules.

This article proceeds as follows. In section 2, we outline the three observationally adequate grammars mentioned above: one grammar in which [ŋ] is derived from /ng/, a second grammar in which [ŋ] is derived from /h/, and a third in which /ŋ/ is an underlying consonant. In section 3, we demonstrate by calculation that under naive representational assumptions, SIMPLICITY favors the descriptively incorrect /h/ → [ŋ] grammar. In section 4, we discuss properties of the /h/ → [ŋ] challenge that make it an instance of a more general problem for phonological learning. This discussion motivates our choice of solution, which does not rely on phonetic similarity. In section 5, we present the solution just outlined, which is based on underspecification and lexical redundancy rules.

2 Three observationally adequate grammars

2.1 /ng/ → [ŋ] grammar (correct)

In English, both [h] & [ŋ] and [n] & [ŋ] are in complementary distribution. The environments in which all three segments surface are shown in (6).

(6) The complementary distributions of [h] & [ŋ] and [n] & [ŋ] in English (cf. Smith 1982; Steriade 2023; C represents any consonant other than [g] and [k])[3]

	[h]	[ŋ]	[n]
#__	✓	✗	✓
$\_\_\begin{Bmatrix} \text{g} \\ \text{k} \end{Bmatrix}$	✗	✓	only as a part of the prefix *un-*
__C	✗	only preceding a deleted underlying /g/	✓
__#	✗	only preceding a deleted underlying /g/	✓
V__V	✓	only preceding a deleted underlying /g/	✓

The distribution of the three segments in (6) licenses the construction of three observationally adequate grammars, which we will describe in this section. First, a grammar in

[3] There are several possible accounts for the surprising appearance of [n], rather than [ŋ], before velar stops in the prefix *un-*. One possibility is that the prefix *un-* belongs to a different cycle than prefixes like *in-* and *con-* (an idea consistent with the observations of Siegel 1974, which categorizes *un-* and *con-* in different classes of prefixes). Under this account, a place assimilation process targeting nasals can be inactive in the cycle in which *un-* is introduced, but active otherwise (e.g., /in-kwɛst/ → [iŋkwέst] '*inquest*'). Another way to exempt *un-* from place assimilation is the proposal by Chomsky and Halle (1968), according to which different morphemes have distinct boundary types that can trigger different phonological rules. If place assimilation does not apply across word boundaries, then the representation /un#/ would block the process.

which [ŋ] is only derived from /n/ before /g/ and /k/; second, a grammar in which [ŋ] is also derived from /h/ in coda position; and third, a grammar in which [ŋ] is derived from /ŋ/. For clarity, grammars are written within the framework of rule-based phonology but the implications should hold for Optimality Theory (OT; Prince and Smolensky 1993/2004) as well.[4]

Let us start with the /ng/ → [ŋ] grammar. In this grammar, every [ŋ] is derived from an underlying /n/ that precedes an underlying /g/ or /k/, as can be seen in the mappings from underlying representations (URs) to surface representations (SRs) in (7). This assimilation pattern is seen transparently in (7a)-(7b) and opaquely in (7c)-(7d), where the trigger for assimilation (/g/) is dropped by a deletion process that targets voiced stops in coda position after nasals.

(7) /ng/ → [ŋ] grammar: lexicon and mappings

a. /in-kwɛst/ → [iŋkwɛ́st] *'inquest'*
b. /fingər/ → [fíŋgər] *'finger'*
c. /sing/ → [síŋ] *'sing'*
d. /inglænd/ → [íŋlænd] *'England'*
e. /bɔmb/ → [bɔ́m] *'bomb'*
f. /hat/ → [hát] *'hat'*
g. /omaha/ → [ómahà] *'Omaha'*
h. /ɛks-heɪl/ → [ɛgzéɪl] ∼ [ɛgzhéɪl] *'exhale'*
i. /ʋihɪkl/ → [ʋíɪkl] ∼ [ʋíhɪkl] *'vehicle'*

The example in (7e) shows that post-nasal stop deletion also applies to voiced labials, as in the mapping of /bɔmb/ → [bɔ́m] (such a process does not apply to voiced coronals, e.g., [ɛ́nd] '*end*'). All other examples show the mapping of an underlying /h/, which can undergo an optional h-dropping process in some environments, as shown in (7h)-(7i).

The phonological rules that account for the alternations in (7) are formalized in (8). The nasal-place assimilation rule is presented in (8a). The rule that deletes voiced dorsals and labials after nasals is presented in (8b). The optional rules in (8c)-(8d) account for the h-dropping in (7h) and (7i) respectively.

(8) /ng/ → [ŋ] grammar: phonological rules

a. $\text{n} \rightarrow [\alpha\ \text{place}]\ /\ \_[\alpha\ \text{place}]$
b. $[\text{-coronal, +voice}] \rightarrow \emptyset\ /\ [\text{+nasal}]\_]_{\sigma}$
c. $\text{h} \rightarrow \emptyset\ /\ \text{C}\_$ (optional)
d. $\text{h} \rightarrow \emptyset\ /\ \_\ \underset{\text{[-stress]}}{\text{V}}$ (optional)

[4]As we will see in section 3, the shorter underlying representations of the /h/ → [ŋ] grammar are beneficial in terms of SIMPLICITY because they extensively outweigh the additional complexity introduced by an /h/ → [ŋ] phonological rule. In OT as well, a cheaper lexicon will push the learner to make the same choice. If constraints are learned, the small counterweight of the rule will be translated into a similarly negligible counterweight of a constraint. If all constraints are innate, there is no counterweight against the lexicon to begin with, since all constraint rankings are equally simple. See Rasin and Katzir (2016, 2020) for relevant discussion.

The derivations in (9) show that the /ng/ → [ŋ] grammar is observationally adequate in the sense that the lexical representations in (7), when combined with the rules in (8), lead to the correct surface forms.[5]

(9) /ng/ → [ŋ] grammar: derivations

UR	/in-kwɛst/	/fingər/	/sing/	/bɔmb/	/ɛks-heɪl/	/ʋihɪkl/
n → [α place] / _[α place]	iŋkwɛst	fiŋgər	siŋg	-	-	-
[-coronal, +voice] → $\emptyset$ / [+nasal]_]$_\sigma$	-	-	siŋ	bɔm	-	-
h → $\emptyset$ / C_ (opt.)	-	-	-	-	ɛkseɪl	-
h → $\emptyset$ / _ $\underset{\text{[-stress]}}{\text{V}}$ (opt.)	-	-	-	-	-	ʋiɪkl
SR	[íŋkwɛst]	[fíŋgər]	[síŋ]	[bɔ́m]	[ɛgzéɪl]	[ʋíɪkl]

2.2 /h/ → [ŋ] grammar (incorrect)

We now turn to the /h/ → [ŋ] grammar, which constitutes the primary competitor for the correct grammar from the perspective of SIMPLICITY. As can be seen in (10a), in this grammar as well, some [ŋ]'s that alternate with [n] must be derived from underlying /n/. Since nasal place assimilation is necessary in this grammar as well, the [ŋ] in (10b) can be derived from either /n/ or /h/. This choice has no effect on the evaluation by SIMPLICITY. The crucial difference between this grammar and the grammar in 2.1 is shown in (10c)–(10d) (in boldface). In those examples, deriving [ŋ] from /h/ rather than from the sequence /ng/ serves an economical purpose, since storing one segment is cheaper than storing two.

(10) /h/ → [ŋ] grammar: lexicon and mappings

a. /in-kwɛst/ → [íŋkwɛst] *'inquest'*
b. /fingər/ → [fíŋgər] *'finger'*
c. **/sih/ → [síŋ] *'sing'***
d. **/ihlænd/ → [íŋlænd] *'England'***
e. /bɔmb/ → [bɔ́m] *'bomb'*
f. /hat/ → [hát] *'hat'*
g. /omaha/ → [ómahà] *'Omaha'*
h. /ɛks-heɪl/ → [ɛgzéɪl] ∼ [ɛgzhéɪl] *'exhale'*
i. /ʋihɪkl/ → [ʋíɪkl] ∼ [ʋíhɪkl] *'vehicle'*

All other examples show no difference from the grammar in 2.1. As can be seen in (10e), post-nasal deletion in coda position must play a role in this grammar as well, even if [ŋ]'s in coda position are derived from /h/.

The phonological rules of this grammar are shown in (11). All four rules from (8) are required here as well. The only difference between the grammars in terms of rules is the additional rule (11e) (in boldface), which is responsible for the derivation of [ŋ] from /h/ in coda position.

[5]The example in (7d) – [iŋlænd] – is derived in the same way as (7c) – [síŋ]. None of the rules in (8) can apply in the derivations of the examples in (7f)-(7g). They are therefore absent from (9) as well. For simplicity, we do not show additional rules that are not relevant to this paper, such as stress assignment and the voicing of the /ks/ sequence in /ɛks-heɪl/ → [ɛgzéɪl].

(11) /h/ $\rightarrow$ [ŋ] grammar: phonological rules

a. n $\rightarrow$ [α place] / _[α place]

b. [-coronal, +voice] $\rightarrow \emptyset$ / [+nasal]_]$_\sigma$

c. h $\rightarrow \emptyset$ / C_ (optional)

d. h $\rightarrow \emptyset$ / _ $\underset{\text{[-stress]}}{\text{V}}$ (optional)

e. **h $\rightarrow$ ŋ / _]$_\sigma$**

The derivations in (12) show that the /h/ $\rightarrow$ [ŋ] grammar is observationally adequate as well. Therefore, this grammar competes with the /ng/ $\rightarrow$ [ŋ] grammar from the perspective of the learner, to which we turn in the next section.

(12) /h/ $\rightarrow$ [ŋ] grammar: derivations

UR	/in-kwɛst/	/fingər/	**/sih/**	/bɔmb/	/ɛks-heɪl/	/ʋihɪkl/
n $\rightarrow$ [α place] / _[α place]	iŋkwɛst	fiŋgər	-	-	-	-
[-coronal, +voice] $\rightarrow \emptyset$ / [+nasal]_]$_\sigma$	-	-	-	bɔm	-	-
h $\rightarrow \emptyset$ / C_ (opt.)	-	-	-	-	ɛkseɪl	-
h $\rightarrow \emptyset$ / _ $\underset{\text{[-stress]}}{\text{V}}$ (opt.)	-	-	-	-	-	ʋiɪkl
h $\rightarrow$ ŋ / _]$_\sigma$	-	-	**siŋ**	-	-	-
SR	[íŋkwɛst]	[fíŋgər]	**[síŋ]**	[bɔ́m]	[ɛgzéɪl]	[ʋíɪkl]

Before turning to the final grammar, we go over the appearance of [ŋ] in morphologically complex words, to confirm that the /h/ $\rightarrow$ [ŋ] grammar is observationally adequate. Most morphologically complex words like '*singer*', '*wrongly*' and '*ring-less*', are pronounced with [ŋ] (and not [ŋg]). In such words, the rule in (11e) applies opaquely, since [ŋ] does not surface in coda position; nonetheless, these words can still be derived from /h/-final URs. The over-application of the /h/ $\rightarrow$ [ŋ] rule in '*singer*' and similar words follows straightforwardly under a cyclic derivation in which vowel-initial suffixes join the derivation after the rule applies to the bare stem, as shown in (13).

(13) A cyclic derivation of '*singer*' in the /h/ $\rightarrow$ [ŋ] grammar:
sih $\xrightarrow[\text{Phonology}]{}$ siŋ $\xrightarrow[\text{Morphology}]{}$ siŋ-er $\xrightarrow[\text{Phonology}]{}$ [siŋer]

However, there are three adjectives that cannot be derived this way, namely '*strong*', '*long*', and '*young*' (Smith, 1982). In the bare form of these adjectives, as well as in affixed forms like '*longish*', [ŋ] is not followed by [g]. Nevertheless, a [g] surfaces in their comparative and superlative forms (e.g., [lɔŋ] vs. [lɔŋg-ər]). In the /h/ $\rightarrow$ [ŋ] grammar, these adjectives must be derived with underlying /ng/, as in '*finger*' in (12). As we will show in the next section, SIMPLICITY's preference for the incorrect grammar is by a large margin, and a handful of words requiring underlying /ng/ cannot tip the scales.

Both in the /h/ $\rightarrow$ [ŋ] grammar and in the /ng/ $\rightarrow$ [ŋ] grammar, these three adjectives can be derived in the same way. In their bare forms, the final /g/ can be deleted by the rule in

(11b) word-finally. In '*longish*', (11b) becomes opaque due to the affixation of -*ish* (cf. 13). The same rule should not apply when the comparative or superlative suffix attaches to the stem. As shown by Bermúdez-Otero (2011), this is predictable if different affixes belong to different cycles. Briefly, the comparative and superlative suffixes enter the derivation before the application of (11b) and bleed it, while suffixes such as -*ish* enter the derivation in later cycles, after the application of (11b). Alternatively, as mentioned in ft. 3, it has been proposed that these suffixes have a different type of boundary than other suffixes, one that does not trigger (11b). See Smith's (1982, p. 394) discussion of Chomsky and Halle's (1968)'s proposal. The particular theory of the morphology–phonology interface one adopts does not affect the challenge posed by the distribution of /h/ and [ŋ], nor the solution we propose in section 5.

2.3 /ŋ/ → [ŋ] grammar (incorrect)

Finally, let us consider the grammar in which [ŋ] is derived from an underlying /ŋ/. As shown in (14), in this grammar as well, some words with [ŋ]'s can be derived from a single segment (/ŋ/), rather than two. We will not spell out the rules and derivations of this grammar because its observational adequacy is straightforward – this grammar includes strictly fewer alternations than the previous two.

(14) /ŋ/ → [ŋ] grammar: lexicon and mappings

a. /in-kwɛst/ → [íŋkwɛst] *'inquest'*
b. /fiŋgər/ → [fíŋgər] *'finger'*
c. **/siŋ/ → [síŋ] *'sing'***
d. **/iŋlænd/ → [íŋlænd] *'England'***
e. /bɔmb/ → [bɔ́m] *'bomb'*
f. /hat/ → [hát] *'hat'*
g. /omaha/ → [ómahà] *'Omaha'*
h. /ɛks-heɪl/ → [ɛgzéɪl] ∼ [ɛgzhéɪl] *'exhale'*
i. /ʋihɪkl/ → [ʋíɪkl] ∼ [ʋíhɪkl] *'vehicle'*

The /ŋ/ → [ŋ] grammar has been defended by Smith (1982), but has been dismissed by others. Sapir (1925) was early to argue that such a grammar is not descriptively adequate and that [ŋ] is not included in the underlying inventory of English. He mentioned the intuition of speakers that [ŋ] "*feels like*" [ng] and pointed out the restricted distribution of [ŋ] before velar consonants (Sapir, 1925, p. 49). In the next section, we will mention a couple of additional arguments from Chomsky and Halle (1968); Fromkin (1973), and Hyman (1975) that support the assumption that [ŋ] is indeed derived from /ng/. If this is true, SIMPLICITY must favor the /ng/ → [ŋ] grammar not just over the /h/ → [ŋ] grammar, but also over the /ŋ/ → [ŋ] grammar, which is also seemingly simpler than the correct grammar.

As mentioned in the introduction, however, there is an independently motivated solution that can make the /ŋ/ → [ŋ] grammar disfavored by SIMPLICITY – the inclusion of CURs. As we will see in section 3, there is a massive SIMPLICITY pressure to minimize the underlying inventory of a language. We will elaborate on this later, but briefly, the size of the underlying inventory determines the encoding cost of each segment in the lexicon, and therefore adding a segment to the underlying inventory inflates the cost of *every* segment in *every* lexical item. The predictability of [ŋ], which can be derived from other underlying segments (/ng/ or /h/), rules out the possibility that SIMPLICITY will choose the /ŋ/ → [ŋ] grammar, provided that CURs are permitted. See Rasin and Katzir (2020) for further discussion. In section 5, we will return to the /ŋ/ → [ŋ] grammar and discuss the circumstances under which SIMPLICITY favors it. Since the solution ruling out the /ŋ/ → [ŋ] grammar has already been identified for a different challenge, the following section, which evaluates the competing grammars, is concerned primarily with the /ng/ → [ŋ] and the /h/ → [ŋ] grammars.

3 Descriptive adequacy and the challenge of explanatory adequacy

As shown in section 2, there are three competing grammars that are consistent with the English data available to the learner: an /ng/ → [ŋ] grammar, an /h/ → [ŋ] grammar, and an /ŋ/ → [ŋ] grammar. There are reasons to believe, however, that only the /ng/ → [ŋ] grammar is descriptively adequate. Supporting evidence can be found in slips of tongue reported in Fromkin (1973, p. 250), in which English speakers seem to pronounce the underlying /g/'s, from which [ŋ] is derived. A few of them are presented in (15).

(15) Slips of tongue that support the /ng/ → [ŋ] grammar (Fromkin, 1973)

	Slip	Target	
a.	[swɪn] and [swej**g**]	[swɪŋ] and [swej]	'*swing and sway*'
b.	[rɪŋk] her [nɛ**g**]	[rɪŋ] her [nɛk]	'*wring her neck*'
c.	[t͡ʃʌŋk jʌ**g**]	[t͡ʃʌk jʌŋ]	'*Chuck Young*'

Another indirect evidence for the /ng/ → [ŋ] grammar is mentioned by Chomsky and Halle (1968, p. 171, fn. 10), who noticed that while there are words like '*line*', '*lime*', '*lane*', and '*lame*' in English, similar nonce words such as *[laɪŋ] and *[leɪŋ] are ungrammatical. According to Chomsky and Halle, this fact can be explained by the independent requirement that diphthongs in English cannot precede a sequence of consonants (e.g., *[baɪlt], *[beɪrd], etc.). This requirement can explain the ill-formedness of words such as [leɪŋ], in which [ŋ] follows a diphthong, as long as [ŋ] is derived from /ng/.

Likewise, Hyman (1975, p. 74) points out that while '*neat*' and '*meat*' exist, *[ŋit] and similar nonce words are systematic gaps. He explains the ungrammaticality of onset [ŋ]'s using an independent requirement of English, according to which there cannot be onset clusters of nasals followed by stops (e.g., *[ŋgit], *[mbit], *[ndit], etc.). If all [ŋ]'s are derived from /ng/ sequences and /ng/ is an impossible onset cluster, the ungrammaticality of *[ŋit] can be predicted.

As we will show shortly, under naive representational assumptions, SIMPLICITY favors the descriptively incorrect /h/ $\rightarrow$ [ŋ] grammar. This wrong prediction of SIMPLICITY constitutes a challenge from the perspective of explanatory adequacy that we are about to address. To understand the nature of this challenge, we first recall how the evaluation metric operates within the learning setting of generative linguistics.

The generative learning setting was outlined by Chomsky (1964, 1965) and is illustrated in (16). In this setting, the evaluation metric is a function that assigns a numerical score to each observationally adequate grammar in the child's hypothesis space. The grammar with the best score is acquired.

(16) A schematic illustration of the learning setting (cf. McCarthy, 1981)

$\boxed{\text{Primary Linguistic Data}} \longrightarrow \boxed{\text{Observationally Adequate Grammars}} \xrightarrow[\text{Metric}]{\text{Evaluation}} \boxed{\text{Descriptively Adequate Grammar}}$

The original evaluation metric for phonological learning is the SPE metric (Chomsky and Halle, 1968). According to this metric, if G and G' are both observationally adequate with respect to the data D, and G is simpler than G', the learner should prefer G to G'. It has been shown, however, that the SPE metric makes incorrect predictions for the acquisition of certain phonological processes, as economy alone can lead to overgeneralization. Specifically, Dell (1981) showed that the SPE metric can motivate the acquisition of grammars that fail to reflect the restrictiveness of certain optional phonological processes.

Simplicity-based learning has since become central again in the form of MDL – an evaluation metric that follows the proposals of Solomonoff (1964), Kolmogorov (1965) and Chaitin (1966). MDL's evaluation is determined by the economy of the grammar ($|G|$; cf. the SPE metric) but also by the economy of the description of the data given the grammar ($|D{:}G|$), which motivates restrictiveness. A statement of MDL is given in (17).

(17) MDL evaluation metric (Rasin et al., 2021):
If G and G' can both generate the data D, and if $|G| + |D{:}G| < |G'| + |D{:}G'|$, prefer G to G'.

MDL has shown success in supporting the acquisition of URs and morphological segmentation from distributional data, even in the face of notable learning challenges such as opacity and optionality. For a comparison of MDL with other phonological learners, see Rasin and Katzir (2016) and Rasin et al. (2021).

The $|D{:}G|$ part of MDL will not play a role in what follows because the competing grammars from section 2 are restrictive to the same extent as far as the calculation of $|D{:}G|$ is concerned.[6] What matters to our discussion is only the cost of URs and phonological rules. Hence, we refer to both MDL and the SPE metric with the term SIMPLICITY that covers all simplicity-based evaluation metrics.

[6]Measuring $|D{:}G|$, however, is crucial for MDL's success with respect to previous known challenges for the SPE metric, namely, the challenge noted by Dell (1981) from optional phonological processes (see Rasin and Katzir 2016; Rasin et al. 2021).

Let us turn to the challenge. As shown above, under naive representational assumptions, the wrong grammars have cheaper lexicons (URs such as /sih/ or /siŋ/ consist of strictly fewer segments than the equivalent /sing/ in the correct grammar). The /h/ $\rightarrow$ [ŋ] grammar, however, has more expensive rules since it includes an additional rule: h $\rightarrow$ ŋ / _]$_\sigma$. We will compare the grammars, translating their differences into a description length in bits, as schematically illustrated in (18). The grammar that can save more bits is favored by SIMPLICITY.

(18) $\underbrace{\underbrace{0101011010100\cdots}_{\text{Lexicon}}\ \underbrace{10101010010\cdots}_{\text{Rules}}}_{\text{G}}$

In Rasin et al. (2021), a phonological rule costs a few dozen bits – an estimation we adopt here without calculating the exact cost of the /h/ $\rightarrow$ [ŋ] rule. As we will show shortly, SIMPLICITY's preference for the incorrect /h/ $\rightarrow$ [ŋ] grammar would persist even if its additional rule cost substantially more than estimated. Since SIMPLICITY is sensitive to the length of URs, we must fix the cost of an underlying segment in order to compare the grammars. Following Rasin and Katzir (2016), the cost of a symbol in the lexicon is the amount of information required to identify it within the set of possible symbols. The formula is given in (19) and the cost for English is calculated in (20). There are other conceivable ways to evaluate the cost of URs, but we follow Rasin and Katzir's (2016) in adopting a naive segment-based approach. We revisit this choice in section 5 as part of our proposed solution.[7]

(19) The cost of a symbol in the lexicon
$\log_2$ x, where x $=$ number of symbols that can be used in the lexicon.

(20) The cost of an underlying segment in English[8]

- There are 51 underlying symbols in English. Therefore, x in English $= 51$.
- The cost of an underlying segment $= \log_2$ x $= \log_2 51 = 5.672$.

Note that the cost in (20) is relatively robust with respect to the exact number of underlying segments in English. Even if there are actually fewer underlying segments, the

[7] An alternative, feature-based cost would not lead to the selection of the correct /ng/ $\rightarrow$ [ŋ] grammar, as long as segments are fully specified in the lexicon. Under this view, segments have a uniform cost (the cost of a feature multiplied by the number of features per segment), which still yields a significant advantage for the shorter URs of the incorrect grammars. As we will see in section 5, however, a feature-based approach in which the cost of an underlying segment varies with the size of the underlying inventory can make the correct predictions. For now, we adopt Rasin and Katzir's (2016) formula, which at least suffices to rule out one of the incorrect grammars.

[8] Our assessment here is based on McMahon's (2016) count of segments in English, which tallies at 50. However, McMahon counts /ŋ/ as well, which should not be considered a lexical symbol in our study under the assumption that [ŋ] is not part of the underlying inventory. We added morpheme and word boundaries to McMahon's count of segments because they also play a role in phonological processes and serve as lexical symbols. Therefore, we set the number 51 for our calculation.

logarithmic function would result in a roughly similar cost for each segment (for example, $\log_2 40 = 5.322$). Such small differences are negligible in the calculation that we are about to see, which concerns only a tiny portion of the entire lexicon.

However, according to this formula, adding a segment to the underlying inventory is translated into an additional cost of all the segments across the lexicon, which includes many words beyond those we are about to count. As long as CURs are available and it is possible to exclude segments whose surface realizations can be derived from other segments, (19) creates immense pressure to shrink the underlying inventory accordingly. Segments like [ŋ], which can be derived from either /ng/ or /h/, are therefore prime candidates for exclusion from the lexicon.

This pressure has consequences that extend beyond the present competition between grammars. As Rasin and Katzir (2020) independently argued, CURs are in fact essential for the learnability of speakers' knowledge of systematic gaps in their language. Although the original argument concerns the distribution of stridents in Dutch, it translates directly to the distribution of [ŋ] in English. Simplifying, while distributional evidence can support the acquisition of a rule changing /n/ to [ŋ] before velars (or /h/ to [ŋ] in coda positions), a rule changing onset /ŋ/ to [n] (or [h]) would be a redundant complexity from the perspective of SIMPLICITY and therefore cannot be acquired, leaving the ungrammaticality of words such as *[ŋit] unpredictable. If CURs are available, however, the formula in (19) motivates the exclusion of /ŋ/ from the underlying inventory altogether. Since no rule then maps any underlying segment to [ŋ] in onset position, *[ŋit] and similar words are correctly ruled out. Rasin and Katzir (2020) show that this provides independent motivation for adopting CURs in general, which in our case exclude underlying /ŋ/.

With /ŋ/ thus eliminated from the underlying inventory, the competition reduces to just two grammars: /h/ $\rightarrow$ [ŋ] and /ng/ $\rightarrow$ [ŋ]. As already noted, the wrong competitor derives [ŋ] in examples such as /sih/ $\rightarrow$ [siŋ] from a single underlying segment, /h/, whereas the correct grammar derives such [ŋ]'s from underlying /ng/ sequences. Consequently, for every word in which [ŋ] is derived from /h/ rather than /ng/, the incorrect grammar requires 5.672 fewer bits – the cost of the additional segment, as determined by the calculation in (20). The total advantage of the incorrect grammar's lexicon is therefore $5.672 \cdot$ y, where y $=$ the number of words containing an [ŋ] derived from underlying /h/. The relevant differences between the grammars are summarized in (21).

(21) The crucial differences between the grammars for SIMPLICITY

	/ng/ $\rightarrow$ [ŋ]	/h/ $\rightarrow$ [ŋ]
Difference in URs	/si**ng/**, /sɔ**ng/**, etc.	/si**h/**, /sɔ**h/**, etc.
Difference in rules		An additional rule: h $\rightarrow$ ŋ / __$]_\sigma$
Advantages in bits	A few dozens (estimated) for one fewer rule	$5.672 \cdot$ y, y $=$ the number of words consisting of an [ŋ] derived from /h/ instead of /ng/

To calculate the advantage of the /h/ $\rightarrow$ [ŋ] grammar, we have searched the Word List corpus of Python's Natural Language Toolkit (NLTK; available online at www.nltk.org), which consists of 236,736 English words. We counted words that end with 'ng', excluding 5-letter or longer words that end with 'ing' (to exclude all verbal present progressive inflections). Our conservative search thus over-excluded relevant words like '*bring*' and '*spring*', and all mid-word coda [ŋ]'s (e.g., '*England*'). The search brought up 417 words, meaning that 2,365 bits can be saved by /h/ → [ŋ] and it is therefore favored over the descriptively correct grammar.

It should be noted that these 417 words include a few duplicated items (e.g., '*king*' and '*King*'), words that are unlikely to be heard by an English learning child (e.g., '*Palaung*' and '*bettong*'), and also the three exceptions mentioned in section 2 ('*long*', '*strong*', and '*young*'). However, even if a more reliable dataset would include just as little as 5% of these 417 words, SIMPLICITY would still favor the incorrect grammar, and fail to address the explanatory adequacy challenge.

4 This is a general challenge

Before turning to our proposed solution, it is worth examining certain properties of the /h/ → [ŋ] challenge that make it general in principle. One might suppose that universal constraints could simply exclude the /h/ → [ŋ] grammar from the child's hypothesis space. Under this view, a universal restriction banning /h/ → [ŋ] alternations would remove this grammar from consideration, leaving SIMPLICITY – assuming CURs – to correctly favor the /ng/ → [ŋ] grammar over the remaining incorrect /ŋ/ → [ŋ] grammar, thereby seemingly dissolving the challenge. Indeed, as mentioned in the introduction, the distribution of [h] and [ŋ] long ceased to be perceived as a challenge precisely on these grounds: the two segments were deemed too phonetically dissimilar to alternate with each other (Hyman, 1975, p. 64). On such a view, a grammar encoding such an alternation should be unavailable. As we will discuss in this section, however, appealing to phonetic similarity to exclude the /h/ → [ŋ] grammar cannot in fact resolve the explanatory adequacy challenge.

We are unaware of any language in which [h] and [ŋ] alternate directly, yet it has been reported that an underlying /g/ in Kalabakan surfaces as [h] intervocalically and as [ŋ] word-finally (Blust, 2016). This pattern undermines any solution that relies on a universal ban on the mapping of /h/ to [ŋ]: since [h] and [g] alternate in Kalabakan, and [ŋ] and [g] alternate as well, such mappings cannot be ruled out universally. Consequently, even if the incorrect /h/ → [ŋ] grammar for English is excluded, a grammar in which /h/ maps to [g] and /g/ maps to [ŋ] must remain in the hypothesis space. Under such a chain-shifting grammar, words like '*song*' can still be derived from a single underlying segment instead of two, albeit from an underlying /g/ instead of /h/ (e.g., /sɔg/ → [sɔŋ]). Words with a coda [g] that does not surface as [ŋ], such as '*dog*' and '*big*', can in turn be derived from an underlying /h/ in coda position (e.g., /dɔh/ → [dɔg]). This chain-shifting grammar requires one more rule than the /h/ → [ŋ] grammar considered so far, since the /h/ → [ŋ] rule is split into /h/ → [g] and /g/ → [ŋ]. However, the calculations in the previous section suggest that the lexical savings should significantly outweigh the addition of one rule. As just discussed, our search brought up 417 words that can benefit from a shorter UR, and roughly 5% are sufficient for justifying the addition of a phonological rule.

There is another reason we will not pursue a solution that rules out specific phonological alternations. We believe that the /h/ → [ŋ] challenge requires a general solution, as a similar problem can arise in principle whenever the three conditions in (22) hold.

(22) General properties of the /h/ → [ŋ] challenge:

a. A trigger of a phonological process is deleted (henceforth, *counterbleeding-by-deletion*, as in /ng/ → |ŋg| → [ŋ], where the trigger /g/ for the place assimilation of /n/ is deleted).

b. The output segment of the counterbleeding-by-deletion interaction happens to be in complementary distribution with another segment in the language (like [ŋ] and [h]).

c. There are enough instances of this interaction to outweigh an additional complexity of rules/constraints (as in our count of words with [ŋ] that can be derived from /h/).

The opaque interaction in (22a) is very common cross-linguistically. According to Kim's (2024) survey of phonological opacity, counterbleeding-by-deletion constitutes roughly two-thirds of all interactions labeled as counterbleeding in the literature. In 41 out of 65 instances of counterbleeding examined, a phonological process is counterbled by a deletion process (Kim, 2024, p. 121). In principle, at least in some of them, (22b) and (22c) may hold as well.

One potential example of such a combination of properties can be found in Bulgarian. In Bulgarian, velar palatalization and yer-vowel deletion are involved in counterbleeding-by-deletion (Green 2004; Kim 2024). The first process turns /k/ into [t͡ʃ] before front vowels and the second process deletes a yer vowel before a syllable containing a full vowel. The interaction of the processes is shown in the derivation of [mrat͡ʃna] '*dark*.fem' in (23).

(23) Counterbleeding-by-deletion in Bulgarian (cf. Green 2004)[9]

UR	/mrakĭna/
Velar palatalization	mrat͡ʃĭna
Yer deletion	mrat͡ʃna
SR	[mrat͡ʃna]

In [mrat͡ʃna], the vowel that triggers palatalization, ĭ, is deleted and the palatalized velar t͡ʃ surfaces before a sonorant consonant. Incidentally, according to Ignatova-Tzoneva (2019), d͡z never surfaces before sonorant consonants (l, r, m, n) in Bulgarian. Therefore, as shown in (24), in an alternative grammar of Bulgarian, [mrat͡ʃna] can be derived from the UR /mrad͡zna/, which consists of one fewer segment than the UR /mrakĭna/.

[9]We follow Green (2004) here and transcribe the front "yer-vowel" using ĭ.

(24) A derivation of [mrat͡ʃna] ‘*dark*.fem’ in an alternative Bulgarian grammar

UR	/mrad͡zna/
$\widehat{\text{dz}} \rightarrow \widehat{\text{tʃ}}$ / __ $\underset{[+son]}{\text{C}}$	mrat͡ʃna
SR	[mrat͡ʃna]

Because velar palatalization and yer deletion are independently motivated, the d͡z → t͡ʃ rule in (24) introduces an additional complexity into the alternative grammar. However, much like the /h/ → [ŋ] case, if there are roughly a dozen words in which [t͡ʃ] can be derived from /d͡z/ instead of the longer /kĭ/, the addition of the extra rule would be justified from the perspective of SIMPLICITY. It seems unlikely that the choice between the grammars in (23) and (24) is solely determined by such a small word count.[10]

A similar example can be found in Japanese. According to Kawahara (2017), palatalization of /s/ before /i/ is counterbled in Japanese by a process of high vowel deletion between voiceless consonants. This is demonstrated in the derivation of [ʃka] ‘*deer*’ in (25).

(25) Counterbleeding-by-deletion in Japanese (cf. Kawahara 2017):

UR	/sika/
s → ʃ / __i	ʃika
$\underset{[+high]}{\text{V}}$ → ∅ / [-voice]__[-voice]	ʃka
SR	[ʃka]

In Japanese, there are no underlying onset clusters. Therefore, an alternative observationally adequate grammar of Japanese can be constructed, in which [ʃ] is derived from /s/ before a consonant, as shown in (26). This alternative grammar should be favored by MDL over the correct grammar of Japanese, similarly to the case of /h/ → [ŋ] in English and the Bulgarian example.

(26) A derivation of [ʃka] ‘*deer*’ in an alternative Japanese grammar:

UR	/ska/
s → ʃ / __C	ʃka
SR	[ʃka]

However, the actual status of the property (22a) in Japanese is not straightforward. There are multiple studies on Japanese that disagree with Kawahara’s (2017) vowel deletion. These studies suggest that the high vowel is not actually deleted, but devoiced (see, for example, Beckman and Shoji 1984; Tsuchida 1997; Whang 2016). According to them, [ʃka] is actually pronounced [ʃi̥ka]. If this description is correct, then in order to maintain

[10]More realistically, there should be two additional rules in the alternative Bulgarian grammar. While the yer-vowel that triggers palatalization is unpronounced in the feminine adjectival form [mrat͡ʃna] (as in the neuter [mrat͡ʃno] and plural [mrat͡ʃni]), it is pronounced in the masculine [mrat͡ʃĭn] ‘*dark*’. Therefore, to derive [t͡ʃ] from /d͡z/ without the underlying vowel, an additional rule, which inserts the vowel in the masculine form, is required. Again, such an addition is justified from the perspective of SIMPLICITY if there are enough words that can benefit from simpler URs.

the shorter URs (such as /ska/), the incorrect grammar must include an additional rule that inserts i̥ after ʃ. Nevertheless, as the calculation in section 3 shows, as long as there are enough words like [ʃi̥ka] that can be derived from shorter URs, the savings in the lexicon should outweigh the cost of the additional epenthesis rule.

What makes the Bulgarian and Japanese case studies especially interesting is the phonetic proximity of [d͡z]∼[t͡ʃ] and [s]∼[ʃ]. As discussed above, the alternation of /g/ with [h] and [ŋ] in Kalabakan complicates any attempt to devise a phonetically based mechanism that can rule out the /h/ → [ŋ] grammar. However, even if some ad hoc solution can still solve the case for English, it is highly unlikely to do the same for Bulgarian and Japanese.[11] In fact, both in the correct and the incorrect grammars of Japanese, [ʃ] is derived from the same consonant /s/ – the difference is only in the environments of the alternation.

In addition to the Bulgarian and Japanese examples, it is possible to think of many other plausible languages which manifest the combination of properties in (22). Consider a hypothetical language in which underlying obstruents are only voiceless, as in Hawaiian (see Newbrand 1951; Lyovin et al. 1997/2017). This hypothetical language has intervocalic voicing (e.g, /tap-a/ → [tabá]) and stressless high vowel deletion (e.g., /tani-la/ → [tanlá]) as systematic phonological processes (stress always falls on the final syllable). Finally, in this hypothetical language, as in English, [h] never occurs in coda position.

The generalizations about this language support an analysis according to which words like [tabmá] are derived from underlying /tapima/. However, under the naive segmental representations we have assumed so far, SIMPLICITY would prefer to derive [tabmá] from /tahma/ as long as there are enough words of this sort to support the addition of an /h/ → [b] rule to the grammar. Regardless of the exact number of words like [tabmá], this analysis does not seem plausible. Importantly, in this hypothetical example, no phonetic restriction is likely to rule out the incorrect grammar, because /h/ → [b] alternations are attested – for example, in Nganasan (see Vaysman 2009) – and therefore cannot be banned.

Since a phonetic solution seems unlikely, what is needed is a formal solution. As discussed throughout this section, we believe that the complementary-distribution challenge for phonological learning is general and can go beyond the particular case of [h] and [ŋ]. Even if it is possible that a different mechanism can solve the problem for English, any phonetic restriction will be an ad hoc solution for a specific phonological pattern and not sufficient for the general case. Therefore, the solution we will pursue in section 5 is entirely formal, relying on information-theoretic considerations and having nothing to do with phonetic restrictions. Of course, one can also think of a hybrid solution whereby the naturalness of phonological rules does not completely rule them out, but rather inflates their cost and makes them less favorable. As we will show immediately, our proposal manages to achieve the desired results without such considerations.

[11] One can argue against the possibility of the alternative Bulgarian grammar, on the grounds that no natural constraint can motivate pre-nasal devoicing. However, post-nasal devoicing and other phonological patterns are similarly unnatural and unmotivated, but nevertheless attested (see Beguš 2019). Such restrictions on phonological mappings may be too restrictive and rule out such attested patterns.

5 Approaching a solution: Underspecification and lexical redundancy rules

5.1 Laying the groundwork

As pointed out by Goodman (1943, 1949), simplicity is defined relative to a representational framework. Consider, for example, two alternative representations of numbers: the Arabic numeral system (0, 1, 2, 3, 4, 5, 6, 7, 8 , 9) and the Roman numeral system (I, V, X, L, C, D, M). As shown in (27), if numbers are represented using Arabic numerals, according to a metric that counts letters, thirty-eight (38) is simpler than a thousand (1000). However, if numbers are represented using Roman numerals, the picture changes and a thousand (M) is simpler than thirty-eight (XXXVIII).

(27) Simplicity of numbers is defined relative to a numeric system

Num. system	One thousand	Thirty-eight
Arabic	1000	**38**
Roman	**M**	XXXVIII

In linguistics as well, different theories of representation will make different predictions about acquisition when combined with a simplicity-based learning criterion. This makes it possible to reason about the correct representational system based on what is acquired using SIMPLICITY. Early work in generative linguistics used this kind of argumentation (Halle, 1962; Chomsky, 1965). Recently, this methodology has been revived using MDL (Katzir, 2014; Katzir et al., 2020; Rasin and Katzir, 2020).

In line with this work, the present paper proposes to rethink the /h/ → [ŋ] puzzle as an argument for specific representations rather than as a challenge for SIMPLICITY. In what follows, we set the foundations of the representational assumptions, which were given in (5) and are repeated in (28), that will help us solve the explanatory adequacy challenge. In subsection 5.2, we revise the segment-base calculation from section 3 and show that under the assumptions in (28), SIMPLICITY favors the correct grammar of English. Because the winning grammar in section 3 was the incorrect /h/ → [ŋ] grammar, we focus in 5.2 only on the competition between this grammar and the correct /ng/ → [ŋ] grammar. Then, in subsection 5.3, we reconsider the /ŋ/ → [ŋ] grammar and discuss the circumstances under which SIMPLICITY would select it given the assumptions in (28).

(28) Representational assumptions that can solve the challenge (cf. Chomsky and Halle 1968, pp. 380–389)

1. In every phonological context (e.g., intervocalically, onset, coda, before a strident, etc.), underlying segments can be underspecified for phonological features as long as they are pairwise distinct.
2. Unspecified features are filled by lexical redundancy rules.

The representational assumptions in (28) come from Halle (1959, 1962) and Chomsky and Halle (1968), who proposed that the lexicon consists of an underspecified level of representation that is filled by lexical redundancy rules. Under this proposal, children try to minimize the number of features they store in their lexicon and can add a lexical redundancy rule to their grammar that fills in the missing features. However, because of SIMPLICITY, children only do so when adding such a rule to the grammar leads to savings in the lexicon that outweigh the cost of the rule itself. According to this proposal, SIMPLICITY favors grammars in which segments are *minimally specified*, given the *minimal set of features* that can generate the data, provided that the savings in the lexicon outweigh the cost of the added lexical redundancy rules.

Halle (1962, p. 60) and Chomsky and Halle (1968, p. 381) made their argument for the assumptions in (28) using the following example. English speakers judge the nonce word *[vnig] as unacceptable but the nonce word [bɪk] as acceptable. For the first word, there is a generalization over the English lexicon that if the second consonant in a word-initial cluster is a nasal or an obstruent, the first consonant must be /s/. The predictability of /s/ in words such as '*snake*' and '*skin*' allows storing the first consonant of those words simply as [+consonantal] and filling in the remaining features of /s/ by a rule like (29), thus saving multiple features in the lexical representation of many words in the lexicon.

(29) [+consonantal]→[+strident, -voice, etc.] / # ___ [+consonantal, etc.]

The rule in (29) bans *[vnig] (and many other ungrammatical nonce words like *[tsajm], *[kbait], etc.) because it requires the consonant preceding /n/ to be /s/.

Unlike *[vnig], [bɪk] differs minimally from English words like '*pick*', '*buck*', and '*big*'. Therefore, a rule that bans [bɪk] but not those words is limited in the number of features it can reduce from the lexicon. Consider, for example, a rule stating that a dorsal stop (/k/ or /g/) in the context [#bɪ_#] must be voiced (/g/). This rule could only save one feature, [+voice], from the lexical entry of '*big*', and therefore, according to Chomsky and Halle, will not be acquired.

Since Halle (1959, 1962) and Chomsky and Halle (1968), there has been much discussion on the nature of underspecification, and alternative theories have been proposed in various frameworks (e.g., Kiparsky 1982, 1993; Archangeli 1988; Steriade 1995; Rasin 2025; see Dresher and Hall 2020 for a comprehensive discussion). In what follows, we show that under Chomsky and Halle's (1968) theory of underspecification, the correct /ng/ → [ŋ] grammar is simpler than the incorrect /h/ → [ŋ] grammar, and thus preferred by SIMPLICITY. While we do not discuss alternative theories of underspecification, an interesting question for future study is whether different theories of underspecification differ in their ability to address the /h/ → [ŋ] challenge.

The preference of SIMPLICITY for the correct /ng/ → [ŋ] grammar is motivated by a generalization that can only be stated in that grammar and can be used to simplify it:

(30) There are no /h/'s in coda position across the English lexicon.

While the correct /ng/ → [ŋ] grammar is consistent with this generalization, in the incorrect /h/ → [ŋ] grammar, there are underlying /h/'s in codas that are used to derive words like '*sing*' and '*song*' (e.g., /sɔh/ → [sɔŋ]).

To understand how the representational assumptions in (28) allow incorporating (30) into the grammar, consider first the toy feature tables in (31) and (32). In order to distinguish between three segments, /n/, /g/, and /k/, as shown in (31), it is possible to use just two features (here chosen to be [±sonorant] and [±voice] for demonstration). However, once we also add /h/ to the inventory, as shown in (32), /k/ is no longer distinguishable by those two features alone. To distinguish between /k/ and /h/, /k/ must be specified for an additional feature (here chosen to be [±continuant]).

(31) Minimally specified /n/, /g/, and /k/ (without /h/)

	n	g	k
±sonorant	+	-	-
±voice	+	+	-
±continuant			

(32) Minimally specified /n/, /g/, /k/, and /h/

	n	g	k	h
±sonorant	+	-	-	-
±voice	+	+	-	-
±continuant			-	+

According to the first assumption in (28), segments must be pairwise distinct, i.e., every segment must have a unique specification that distinguishes it from every other segment. Crucially, a specification that is a proper subset of another segment's specification does not count as unique (see Chomsky and Halle 1968, pp. 382-389 for further discussion of pairwise distinctness, building on the observations of Lightner 1963 and Stanley 1967). Consequently, adding /h/, which is [-sonorant, -voice, +continuant], to the underlying inventory forces /k/, which shares the features [-sonorant, -voice], to be specified as [-continuant] in order to remain distinct from /h/.

Therefore, if underspecification is allowed and there are no /h/'s in coda position, as in the correct grammar, the exclusion of underlying /h/'s from coda position permits a substantial reduction in the number of features stored in the lexicon. Because of the exclusion of coda /h/'s, the features that distinguish between /k/ and /h/ can be removed from coda /k/'s in the lexicon and filled in by a lexical redundancy rule. In what follows, we will revise the calculation from section 3 and show that, under the current representational assumptions, SIMPLICITY correctly favors the /ng/ → [ŋ] grammar.

5.2 A feature-based revision of the calculation from section 3

Under the architecture in (28), we can revise the segment-based calculation in (21) and count features in the lexicon rather than segments. For a concrete illustration, consider the exhaustive feature table in (33). The table shows all fully specified underlying English consonants, assuming the SPE theory of features.[12]

[12]We use some common abbreviations here and below: son = sonorant, voc = vocalic, cons = consonantal, cor = coronal, ant = anterior, cont = continuant, and str = strident. Chomsky and Halle (1968) also use the features [±tense] and [±round], both unspecified for all consonants in (33) except /k/ and /g/, which are both [-round].

(33) Fully specified English consonants

	p	b	f	v	m	t	d	θ	ð	s	z	ʃ	ʒ	t͡ʃ	d͡ʒ	n	r	l	k	g	j	w	h
±son	-	-	-	-	+	-	-	-	-	-	-	-	-	-	-	+	+	+	-	-	+	+	-
±voc	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	+	-	-	-	-	-
±cons	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-	-
±cor	-	-	-	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-	-	-	-
±ant	+	+	+	+	+	+	+	+	+	+	+	-	-	-	-	+	-	+	-	-	-	-	-
±voice	-	+	-	+	+	-	+	-	+	-	+	-	+	-	+	+	+	+	-	+	+	+	-
±cont	-	-	+	+	-	-	-	+	+	+	+	+	+	-	-	-	+	+	-	-	+	+	+
±nasal	-	-	-	-	+	-	-	-	-	-	-	-	-	-	-	+	-	-	-	-	-	-	-
±str	-	-	+	+	-	-	-	-	-	+	+	+	+	+	+	-	-	-	-	-	-	-	-
±back	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	+	-	+	-
±high	-	-	-	-	-	-	-	-	-	-	-	+	+	+	+	-	-	-	+	+	+	+	-
±low	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+

The minimally specified version of the table in (33) is shown in (35). The table was constructed while preserving Rasin and Katzir's (2016) assumption about cost, given in (19) and repeated in (34).

(34) The cost of a symbol in the lexicon
$\log_2$ x, where x = number of symbols that can be used in the lexicon.
Henceforth, $\log_2 \text{x} = c$

Just like the cost formula in (34) created pressure for reducing the number of underlying segments in our previous segment-based calculation, it motivates the minimization of the number of features used in the lexicon under a feature-based calculation. The table in (35) therefore shows the minimal specification of English consonants given the minimal set of distinctive features. This table illustrates the representations of consonants intervocalically, where all of them are available. The specification of the consonants in (35) is approximately the minimum required to distinguish them all from each other.

(35) Minimally specified English consonants

	p	b	f	v	m	t	d	θ	ð	s	z	ʃ	ʒ	t͡ʃ	d͡ʒ	n	r	l	k	g	j	w	h
±son	-	-			+	-	-	-	-	-	-	-	-			+	+	+	-	-	+	+	-
±cor	-	-	-	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-	-	-	-
±ant	+	+	+	+		+	+	+	+	+	+	-	-	-	-	+	-	+	-	-	-	-	-
±voice	-	+	-	+		-	+	-	+	-	+	-	+	-	+				-	+			-
±cont	-	-	+	+	-	-	-	+	+	+	+	+	+	-	-	-	+	+	-		+	+	+
±str								-	-	+	+												
±back																					-	+	

The table in (35) is an approximation of the simplest way to specify the consonants of English under the pressure to minimize the general set of features.[13] For example, /m/

[13]The table makes certain arbitrary choices for concreteness. For example, /j/ and /w/ are specified as

and /n/ could have been defined using [+nasal], but they are specified using [+son, -cont] because [±son] and [±cont] are independently required. A feature is only used if it is necessary. An example is the feature [±str], which is needed to distinguish the coronal fricatives /θ/ and /ð/ from /s/ and /z/.[14] After constructing the table using the minimal set of distinctive features, a secondary pressure is to reduce every redundant feature from the specifications of every segment in the lexicon. Consider /k/ for example: [-son] is required to distinguish it from /m/, [-cor] from /t͡ʃ/, [-ant] from /p/, [-voice] from /g/, and [-cont] from /h/. All other features of /k/ are redundant and can be filled in by lexical redundancy rules.

However, in the correct /ng/ → [ŋ] grammar, there are no /h/'s in codas. The feature [-cont] can be removed from the specification of /k/ in coda position, because there is no other [-son, -cor, -ant, -voice] consonant in the same position in that grammar. This is visualized in the table in (36), which illustrates the representation of English consonants in coda position. It approximates the minimum of features required to distinguish all of the consonants from each other, except /h/, which is excluded from coda position in the /ng/ → [ŋ] grammar.

(36) Minimally specified English consonants **without /h/**

	p	b	f	v	m	t	d	θ	ð	s	z	ʃ	ʒ	t͡ʃ	d͡ʒ	n	r	l	k	g	j	w
±son	-	-			+	-	-	-	-	-	-	-	-			+	+	+	-	-	+	+
±cor	-	-	-	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-	-	-
±ant	+	+	+	+		+	+	+	+	+	+	-	-	-	-	+	-	+	-	-	-	-
±voice	-	+	-	+		-	+	-	+	-	+	-	+	-	+				-	+		
±cont	-	-	+	+	-	-	-	+	+	+	+	+	+	-	-	-	+	+			+	+
±str								-	-	+	+											
±back																					-	+

The crucial difference between (35) and (36) was highlighted in (3) and (4) and is repeated here in (37) and (38). In the incorrect /h/ → [ŋ] grammar, the lexicon must include enough features to distinguish between underlying /h/ and /k/ in coda position. As shown in (37), the feature [±continuant] is required to make /k/ distinguishable from /h/. In the correct grammar, however, /h/ never appears in coda position, so it is redundant to specify the feature [-continuant] for coda /k/, as illustrated in (38).

[-ant] in order for them to be distinguished from /f/ and /v/. Alternatively, we could have specified /f/ and /v/ as [-son]. Because segment frequency affects the aggregated feature cost of each segment, such decisions can influence the overall cost of the lexicon. SIMPLICITY could thus favor one option over the other. In this study, however, we restrict our attention to the feature counts for /n/, /g/, /h/, and /k/, the segments directly relevant to the analysis.

[14]The feature [+str] is redundant for /ʃ/, /ʒ/, /t͡ʃ/, and /d͡ʒ/ in (35), as it is not required to keep these segments pairwise distinct. Nevertheless, [+str] must be present for the application of phonological processes such as epenthesis between stridents (in words like '*busses*', '*bushes*', '*matches*', etc.). Under the current assumptions, a lexical redundancy rule can fill in [+str] for segments such as /ʃ/ and /t͡ʃ/ prior to the application of regular phonological rules. The same logic extends to other consonants in (35) whose unspecified features are required for the application of phonological processes.

(37) Underspecified segments in coda position, **incorrect grammar**

	p	b	...	k	h
±coronal	-	-	...	-	-
±anterior	+	+	...	-	-
...			...		
±voice	-	+	...	-	-
±continuant	-	-	...	-	+

(38) Underspecified segments in coda position, **correct grammar**

	p	b	...	k
±coronal	-	-	...	-
±anterior	+	+	...	-
...			...	
±voice	-	+	...	-
±continuant	-	-	...	

It should be noted that the theory of features plays a critical role in our proposal. As just outlined, the solution relies on the fact that adding /h/ to a given position forces another segment in that position – in our case, /k/ – to take on additional features to remain pairwise distinct. If /h/ could be added to coda position without increasing the feature specifications of other segments in that position, the solution would not go through. In this article, we chose to illustrate our proposal using the SPE theory of features, under which SIMPLICITY correctly favors the /ng/ → [ŋ] grammar. However, this should not be taken as an argument supporting the SPE feature theory specifically, but rather as a proof of concept for our proposal. Whether the choice of feature theory affects the outcome is a question we leave for future research.

Let us proceed to the evaluation of the competing grammars. Unlike the calculation in (21), under the current feature-based assumptions, the cost of an underlying segment is determined by the number of features required to specify it in (35) or (36). The crucial differences in the representation of the grammars are shown in (39), which we use as the basis for the feature cost calculation below. As mentioned before, since the incorrect /ŋ/ → [ŋ] grammar was already disfavored in the previous calculation, we leave it aside for now and return to it in subsection 5.3.

(39) A revision of (21), assuming underspecification.

	/ng/ → [ŋ]	/h/ → [ŋ]
Difference in URs	/si**ng**/, /sɔ**ng**/, etc.	/si**h**/, /sɔ**h**/, etc.
Difference in rules		/h/ → ŋ / $\_]_\sigma$
Advantages	The additional feature required to distinguish between /k/ and /h/, for every coda /k/.	The additional features required to state /ng/ instead of /h/, for every [ŋ].

Note that the lexical redundancy rules themselves do not play a role in this comparison, because the only difference between these grammars in this respect is in the statement of the feature-filling rule for /k/ and, as shown in (40), it has the same complexity in both grammars. The only difference between the two versions of the rule is that the feature [-cont] is stated in the output of the rule in the /ng/ → [ŋ] grammar, but in the input of the rule in the /h/ → [ŋ] grammar.

(40) Feature-filling rule for /k/ in coda position:

- /ng/ → [ŋ] grammar:
  [-son, -cor, -ant, -voice] → [**-cont**, -str, etc.] / $\_\_]_\sigma$
- /h/ → [ŋ] grammar:
  [-son, -cor, -ant, -voice, **-cont**] → [-str, etc.] / $\_\_]_\sigma$

Given the featural representations in (35) and (36), storing /sih/ instead of /sing/ in the lexicon of the /h/ → [ŋ] grammar saves the cost of 7 symbols (3 additional features, their binary value, and an additional separator of segments, which is required to divide the different bundles of features that specify /n/ and /g/). On the other hand, excluding /h/ from codas in the correct grammar saves 2 symbols (the feature [±continuant] and its value) for every /k/ in coda position. Because we only counted word-final [ŋ]'s in section 3, we also count here only words that end with 'k', excluding many medial-word coda /k/'s (e.g., [sí́krɪt] '*secret*').[15] We searched the same corpus of Python's NLTK package and found 2,674 such words. The new evaluation is shown in (41).

(41) A revision of (21), assuming underspecification - **bit calculation**

	/ng/ → [ŋ]	/h/ → [ŋ]
Difference in URs	/si**ng**/, /sɔ**ng**/, etc.	/si**h**/, /sɔ**h**/, etc.
Difference in rules		/h/ → ŋ / $\_\_]_\sigma$
Advantages in bits	A few dozens (estimated) for one fewer rule. 2 symbols for coda /k/ $= 2 \cdot 2,674 \cdot c = 5,345 \cdot c$	7 symbols for each word with [ŋ] derived from /h/ instead of /ng/ $= 7 \cdot 417 \cdot c = 2,919 \cdot c$

As can be seen in (41), the /ng/ → [ŋ] grammar now wins. However, what may still be worrying in the picture portrayed by (41) is the fact that if there were sufficiently more than 417 words that can benefit from shorter URs in the /h/ → [ŋ] grammar, the result would have been reversed. In section 4, we argued that the logical analysis should not be determined by a small word count. A closer examination of the current situation reassures that this is indeed not the case here.

Consider the conditions in which, assuming underspecification, the /ng/ → [ŋ] grammar would be disfavored. In order for the advantages of the /h/ → [ŋ] grammar to outweigh the advantages of the correct grammar, the number of words with [ŋ] derived from /h/ must be almost double than the current 417. Roughly 775 words would have been enough to justify the addition of a feature to coda /k/'s and the /h/ → [ŋ] rule. First of all, this is not a small modification, but a substantial change to the lexicon. However, more importantly, given the current representational assumptions, there is no reason to assume that even the addition of hundreds of words to English will result in [ŋ] being derived from /h/. Under the representational assumptions we started the article with, both the /ng/ → [ŋ] and the

[15]Presumably, there are more medial coda /k/'s than medial [ŋ]'s that can be derived from /h/'s, which would further strengthened the difference in favor of the /ng/ → [ŋ] grammar.

/h/ $\rightarrow$ [ŋ] were simpler than the /ŋ/ $\rightarrow$ [ŋ] grammar, but the current situation, under the assumptions in (28), is different: there is now no clear reason to add /h/'s to coda position, rather than underlying [ŋ]'s, as we explain in the following subsection.

5.3 Back to the /ŋ/ $\rightarrow$ [ŋ] grammar

Under the initial assumptions with which we began, the main simplicity-based pressure was to reduce the number of underlying segments, and therefore both the /h/ $\rightarrow$ [ŋ] and /ng/ $\rightarrow$ [ŋ] grammars were favored over /ŋ/ $\rightarrow$ [ŋ] grammar. As discussed in section 3, the pressure came from the formula in (34), according to which the addition of a segment to the underlying inventory entailed an additional cost for every segment in every lexical item (because the cost of a segment was determined by the number of underlying segments). Under the current feature-based assumptions, however, adding /ŋ/ to codas is considerably less costly. Under the current assumptions, adding distinctive features to the underlying inventory inflates the cost of the entire lexicon. However, as long as adding a segment to a given environment does not require expanding the overall feature inventory, such an addition only affects the specification of one other segment in that environment, much as adding /h/ to codas above required the addition of a feature for coda /k/'s.

Indeed, adding /ŋ/ to the lexicon does not require more features than those used in (35) and (36). Adding /ŋ/ to codas would affect the specification of coda /m/'s only, as /m/ is the only other segment specified as [+son, -cor, -cont].[16] Moreover, deriving [ŋ] from /ŋ/ also avoids the cost of the additional phonological rule required by the /h/ $\rightarrow$ [ŋ] grammar.

In other words, the /ŋ/ $\rightarrow$ [ŋ] grammar, which was the worst option under the previous assumptions, is now preferable over the /h/ $\rightarrow$ [ŋ] grammar according to SIMPLICITY. It follows that the type frequency of words with [ŋ] may influence the competition between the /ng/ $\rightarrow$ [ŋ] and /ŋ/ $\rightarrow$ [ŋ] grammars. The prediction is that an increase in the number of words with [ŋ] would trigger a switch to the /ŋ/ $\rightarrow$ [ŋ] grammar, rather than to the /h/ $\rightarrow$ [ŋ] grammar.

To summarize this section, under the representational assumptions of Halle (1959, 1962) and Chomsky and Halle (1968), SIMPLICITY favors the descriptively correct grammar of English over the /h/ $\rightarrow$ [ŋ] grammar. In the correct grammar, as outlined in section 2, [ŋ] is derived from the sequence /ng/ in coda position. /n/ assimilates to /g/, which is deleted by a process that targets non-coronal voiced stops after nasals (e.g., /sing/ $\rightarrow$ |siŋg| $\rightarrow$ [siŋ]). SIMPLICITY chooses this grammar even though it has longer URs, because this grammar captures the generalization in (30), which can simplify the lexicon elsewhere. The exclusion of /h/ from coda position, assuming underspecification, licenses the removal of the feature [-cont] from all coda /k/'s. This feature is filled in by the lexical redundancy rule given in (40) and repeated here: [-son, -cor, -ant, -voice] $\rightarrow$ [-cont, -str, etc.] / $\_\_]_\sigma$.

[16]Given the table in (36), in order to be distinguishable from /ŋ/, /m/ needs the additional specification of [+ant], as /ŋ/ can be specified as [+son, -cor, -ant, -cont].

6 Conclusion

The complementary distribution of /h/ and [ŋ] was originally presented as a challenge for structuralism. As we have shown, this distribution also poses a broader challenge for phonological learning. Under naive representational assumptions, the descriptively incorrect /h/ $\rightarrow$ [ŋ] grammar is simpler than the descriptively correct /ng/ $\rightarrow$ [ŋ] grammar. A simplicity-based learner therefore favors the incorrect grammar – an explanatory adequacy failure.

Chomsky and Halle (1968) did not seem to worry about this puzzle, even though it seems to challenge the SPE metric. In any case, it could have been set aside for two reasons: a phonetic-based solution seemed within reach, and simplicity-based learning had largely been abandoned. Neither condition holds today. As we argued in section 4, the /h/ $\rightarrow$ [ŋ] challenge is unlikely to be resolved by a phonetic solution. And with the revival of simplicity-based learning in the form of MDL, the challenge can no longer be ignored.

The solution we propose is to reframe the problem not as a challenge for SIMPLICITY per se, but for the combination of SIMPLICITY and certain representational assumptions. Abandoning the assumption that underlying segments are fully specified, as well as abandoning ROTB, dissolves the problem. Crucially, it is SIMPLICITY itself that does the explanatory work: once underspecification and lexical redundancy rules are assumed, the simpler grammar is also the correct one. Whether alternative learning frameworks could handle this challenge at all remains an open question.

What is clear is that Chomsky and Halle were right not to worry: they had independently argued for both ingredients of the solution — the representational assumptions that can eliminate the challenge, and a simplicity-based evaluation metric that selects the correct grammar under these assumptions.

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